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MILITARY SPECIFICATION

WIRE ROPE, STEEL (CARBON) FLEXIBLE, PREFORMED

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1. This specification covers the performed type (see section 6) of flexible carbon-steel cable.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

MM-L-736—Lumber and Timber; Hardwood
MM-L-751—Lumber and Timber; Softwood
QQ-S-781—Strapping, Flat, Steel
QQ-S-790—Steel Strapping, Round, (Bare and Zinc-Coated)
UU-P-271—Paper, Wrapping, Waterproofed Kraft

MILITARY

MIL-B-121—Barrier Material, Greaseproofed, Flexible (Waterproofed)
MIL-C-16173—Corrosion Preventive Compound, Solvent Cutback, Cold Application

STANDARDS

FEDERAL

FED. TEST METHOD STD. No. 151—Metals; Test Methods

MILITARY

MIL-STD-129—Marking for Shipment and Storage

(Copies of documents required by contractors in connection wtih specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The steel used in the manufacture of cable (see section 6) covered by this specification shall be manufactured by the acid-open-hearth, basic-open-hearth, or electric furnace process. Every precaution shall be taken to keep the products of different heats separate and identifiable throughout the manufacturing operations.

3.2 Wire (see section 6). The wire used in construction of flexible steel cable shall be cylindrical and smooth and of uniformly high quality. It shall be free from splits, cold shuts, and other defects. The wire shall be thoroughly and uniformly coated with pure tin or zinc.

3.2.1 *Joining.* Splices in individual wires 0.014 inch in diameter (see section 6) and smaller may be twisted. Joints in wires of larger diameter shall be brazed or welded. Splices in individual wires in any layer of a strand (see section 6) shall be not closer than 20 feet. Brazed joints shall be tinned or zinc coated, but exposed brazing metal shall not constitute cause for rejection.

3.3 Cable construction. The individual wires and strands composing the cable shall be preformed into the exact helical position they will have in the finished cable, so that if the cable is cut or severed, there is no tendency for the normal diameter of the cable at the unseized ends to increase by more than the amount specified in table I.

3.3.1 *Cable lengths.* The cable shall be furnished in minimum lengths of 1,000 feet, except that 20 percent of the cable in the order may be furnished in lengths of 500 to 1,000 feet.

3.3.2 *Lubrication.* The cable shall be impregnated with a suitable friction-preventive

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compound. The amount of compound adhering to the finished cable after the application shall be such that a white cloth, which has been rubbed once over a 1-foot length of the cable, will show a noticeable amount of the compound. The friction-preventive compound shall possess corrosion-resistant properties. (See 6.4.)

3.3.3 *Types of construction.* Flexible steel cable covered by this specification shall be of 3 by 7, 7 by 7, 7 by 19, or 6 by 19 independent wire rope center (IWRC) construction according to the diameter of the cable as specified in table I.

TABLE I.—*Construction and properties*

Nominal diameter of cable (inches)	Construction	Tolerance on diameter (plus only)	Allowable increase in diameter at unseized end (maximum)	Breaking strength of cable (minimum)	Weight per 100 feet (approximate)
		Inch	*Inch*	*Pounds*	*Pounds*
1/32	3 by 7	0.006	0.006	110	0.16
1/16	7 by 7	.010	.009	480	.75
3/32	7 by 7	.012	.010	920	1.60
1/8	7 by 19	.014	.011	2,000	2.90
5/32	7 by 19	.016	.017	2,800	4.50
3/16	7 by 19	.018	.019	4,200	6.50
7/32	7 by 19	.018	.020	5,600	8.60
1/4	7 by 19	.018	.021	7,000	11.00
9/32	7 by 19	.020	.023	8,000	13.90
5/16	7 by 19	.022	.024	9,800	17.30
11/32	7 by 19	.024	.025	12,500	20.70
3/8	7 by 19	.026	.027	14,400	24.30
7/16	6 by 19 (IWRC)	.030	.030	17,600	35.60
1/2	6 by 19 (IWRC)	.033	.033	22,800	45.80
9/16	6 by 19 (IWRC)	.036	.036	28,500	59.00
5/8	6 by 19 (IWRC)	.039	.039	35,000	71.50
3/4	6 by 19 (IWRC)	.045	.045	49,600	105.20
7/8	6 by 19 (IWRC)	.048	.048	66,500	143.00
1	6 by 19 (IWRC)	.050	.050	85,400	187.00
1 1/8	6 by 19 (IWRC)	.054	.054	106,400	240.00
1 1/4	6 by 19 (IWRC)	.057	.057	129,400	290.00
1 3/8	6 by 19 (IWRC)	.060	.060	153,600	330.00
1 1/2	6 by 19 (IWRC)	.062	.062	180,500	420.00

3.3.3.1 *7 by 7 construction.* Cable of this construction shall consist of 6 strands of 7 wires each, laid around a center strand (see section 6) of 7 wires. The wire shall be laid in accordance with the best commercial practice so as to develop a cable which will produce the greatest bending and wearing properties. The cable shall have a length of lay (see section 6) of not more than 8 times nor less than 6 times the cable diameter.

3.3.3.2 *7 by 19 construction.* Cable of this construction shall consist of 6 strands laid around a center strand. The wire composing the 7 individual strands shall be laid around a central wire (see section 6) in two layers. The single core strand shall consist of a layer of 6 wires laid around the central wire in a right-hand direction, and a layer of 12 wires laid around the 7 wire-strand in a right-hand direction. The 6 outer strands of the cable shall consist of a layer of 6 wires laid around the central wire in a left-hand direction, and a layer of 12 wires laid around the 7-wire strand in a left-hand direction.

3.3.3.2.1 *Length of lay.* The length of lay of the layers of both the core strand and outer strands shall be as follows: The outside layer of 12 wires of the strands shall have a length of lay not greater than 50 percent of the cord or cable lay. The inside layer of the strands

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consisting of 6 wires around the central wire shall have a length of lay not greater than 60 percent of the lay of the outside layer of 12 wires.

3.3.3.2.2 The 6 outer strands shall be laid around the core strand in a right-hand direction, with a length of lay not more than 8 nor less than 6 times the cable diameter.

3.3.3.3 *6 by 19 (IWRC).* Cable of this construction shall consist of 6 strands of 19 wires each, laid around a 7 by 7 independent wire rope (see secetion 6) center. The 7 by 7 independent wire rope center shall consist of cable or wire rope of 6 strands of 7 wires each, twisted or laid around a strand center or core consisting of 7 wires.

3.3.3.4 *3 by 7 construction.* Cable of this construction shall consist of 3 strands of 7 wires each. There shall be no core in this construction. The cable shall have a length of lay of not more than 8 times nor less than 5 times the nominal cable diameter.

3.3.3.5 *Physical properties.* Diameter, breaking strength, and weight shall be as specified in table I.

3.4 **Endurance test properties.** Cable of the sizes included in table II shall be capable of withstanding endurance testing for the number of reversals and at the loads required by table II.

3.5 **Breaking strength after endurance test.** The breaking strength of the cable after being subjected to the endurance test specified in 4.6 shall be not less than the values shown in table II.

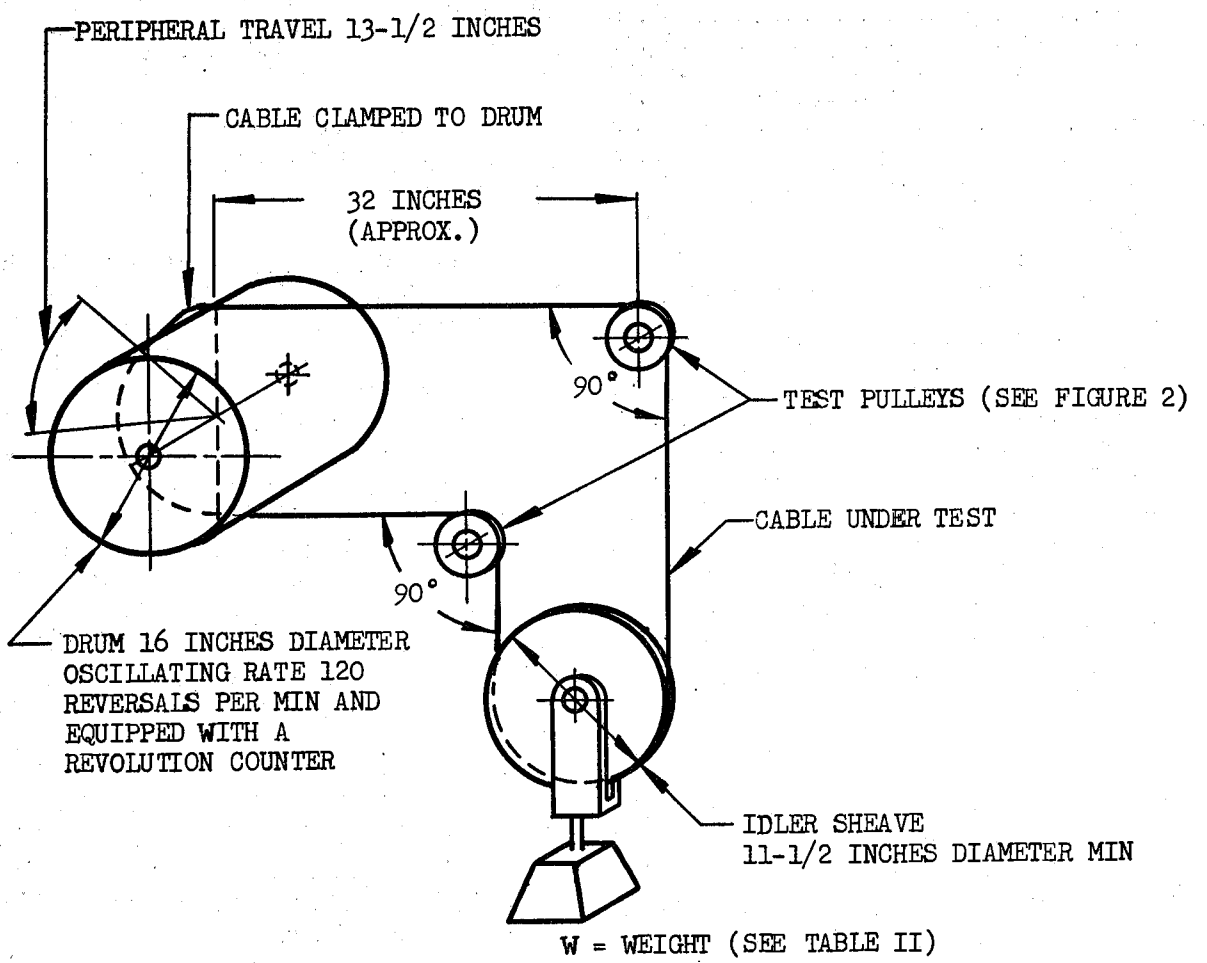


FIGURE 1.—*Cable Endurance Testing Machine (Diagrammatic).*

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TABLE II.—*Endurance test conditions*

Cable diameter (inch)	Tension [1]	Number of reversals	Breaking strength (minimum)
	Pounds		*Pounds*
1/8	20	70, 000	1, 000
5/32	30	160, 000	1, 400
3/16	45	130, 000	2, 100
1/4	70	130, 000	3, 500

[1] Tension is equal to half of weight (W) which includes idler sheave and hanger, as shown in figure 1.

3.6 Wire coating.

3.6.1 *Zinc coating.* Zinc coating of the wires shall be the hot-dipped or electroplated process and shall conform to the following requirements:

Wire diameter (inch)	*Minimum weight of zinc coating (oz/sq ft)*
0.005 to 0.010	0. 03
Over .010 to .015	.05
Over .015 to .028	.10
Over .028 to .060	.20
Over .060 to .090	.30
Over .090 to .140	.40

3.6.2 *Tin coating.* Tin coating of the wires shall be by the hot-dipped or electroplated process and shall conform to the following requirements:

Wire diameter (inch)	*Minimum weight of tin coating (oz/sq ft)*
0.005 to 0.010	0. 003
Over .010 to .015	.005
Over .015 to .028	.010
Over .028 to .0625	.015
Over .0625 to .1205	.020
Over .1205 to .162	.025

3.7 Workmanship. All details of workmanship shall be in accordance with the best practice for high quality aircraft cable, consistent with the requirements of this specification. The cable shall be securely laid and free from kinks, loose wires, loose strands, or other defects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 General. All the tests required herein for the testing of flexible carbon-steel cable are classified as acceptance tests, for which necessary sampling techniques and method of testing are specified in this section.

4.3 Examination of product. All cable shall be examined for workmanship and finish. For this purpose, the cable shall travel no faster than 100 feet per minute and shall be stopped for closer inspection when deemed necessary. This inspection shall take place at the time the cable is wound on shipping reels. Any discard from the head end of the manufacturing reel shall be removed prior to winding on the shipping reels. Protruding wires are considered a defect, and not more than one broken end will be accepted in 1,000 feet of cable.

4.3.1 Shipping reels shall be visually examined for compliance with the requirements of section 5.

4.4 Sampling.

4.4.1 One sample, not less than 40 feet in length for sizes listed in table II, and not less than 15 feet in length for other sizes shall be taken, after any discard has been removed, from the head or starting end of the first manufacturing reel for each lot of cable. In addition to this sample, one sample shall be taken from each shipping reel of 10,000 feet or less, except that for shipping reels of 5,000 feet or less, one sample shall be taken from each 5,000 feet of cable from each lot. A lot shall consist of cable of the same construction and diameter, produced continuously by one machine or one series of progressive processing machines.

4.4.2 Each manufacturing reel shall be given a lot number by the manufacturer. When the manufacturing reel is cut into specified lengths for shipping reels, each shipping reel shall be marked with the lot number and its location in the manufacturing reel, starting from the head end and numbering each shipping reel consecutively.

4.5 Breaking strength.

4.5.1 *Preparation of specimens.* Three

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specimens for determining breaking strength of cable shall be prepared from each sample of cable selected. These specimens shall be straight and not less than 24 inches long.

4.5.2 *Method.* Breaking strength of cable shall be determined on a tension testing machine in accordance with Fed. Test Method Std. No. 151. The distance between the jaws of the machine, with the sample in place ready for testing, shall be not less than 10 inches. Specimens may be clamped in the testing machine jaws in the usual manner by socketing in zinc or other suitable metal or by swaging fittings on ends.

4.5.3 *After endurance test.* The breaking strength after endurance test (4.6) shall be determined as previously outlined herein, except that the specimen for this test shall be so selected as to determine the breaking strength of a portion of the cable that has been subjected to reverse bending by contact with a test pully on a machine or test stand conforming to the requirements of figure 1.

4.6 Endurance test.

4.6.1 *Preparation of specimens.* One endurance test specimen 13 feet long shall be prepared from each sample of cable conforming to diameters shown in table II.

4.6.2 *Method.* Test specimens shall be tested a indicated by figure 1, employing test pulleys conforming to the dimensions of figure 2, to determine compliance with the specified endurance requirements. The number of oscillations per minute, or total of reversals, and the tension in the cable for the endurance test shall be as shown in figure 1 and table II for the respective size of cable. The total travel of the cable in one direction shall be 13½ inches. The application of lubricant other than or in addition to that used to impregnate the cable during the process of manufacture shall not be permitted either before or during the endurance test.

4.7 Weight of tin or zinc coating.

4.7.1 *Preparation of specimens.* Three specimens not less than 12 inches long shall be cut from the wire before the cable is fabricated. These specimens shall be cut from each size of wire to be used in making each master reel of cable.

4.7.2 *Method, tin-coated wire.* The weight of coating on samples of tin-coated wire shall be determined by procedure D, method 511, Fed. Test Method Std. No. 151.

4.7.3 *Method, zinc-coated wire.* The weight of coating on samples of zinc-coated wire shall be determined by the stripping test method of method 511, Fed. Test Method Std. No. 151.

4.8 Rejection and retest. The failure of any specimen shall be cause for the rejection of the lot represented, except that the manufacturer may, at his own expense, in the presence of the Government inspector, sample each length of cable comprising a rejected lot and subject specimens from these samples to retest. The inspector may accept lengths of cable shown to conform to all the requirements of this specification by this retest.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

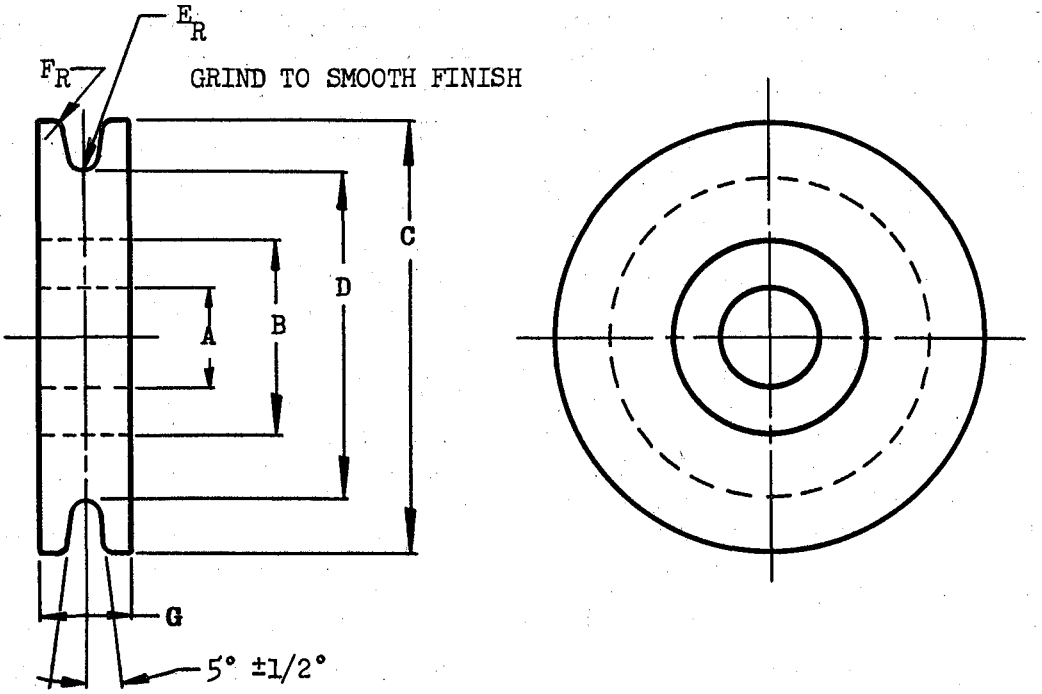
5.1.1 *Preservation—Levels A and C.* All cable shall be coated with a continuous film of corrosion preventive compound conforming to grade 2 of Specification MIL-C-16173, applied by dipping or immersion in a solution of compound and solvent at room temperature and prior to winding on the shipping reel.

5.1.2 *Packaging.* The cable shall be wound, one size on a reel, on reels constructed to the dimensions specified in table III. Before starting to wind the cable on the reel, a layer of waterproof barrier material, Specification UU-P-271, and then a layer of greaseproof paper, Specification MIL-B-121, shall be applied to the barrel of the reel and the inside of the flanges against which the cable will subsequently rest. After the cable is wound on the lined reel, a layer of greaseproof paper, Specification MIL-B-121, and then a layer of waterproof barrier material, Specification UU-P-271, shall be applied and fastened by the use of three or four wooden lagging strips and two metal straps. The metal strapping shall conform to Specification QQ-S-781 or QQ-S-790.

5.2 Packing. Packing shall be level A or C as specified (see 6.2).

5.2.1 *Level A.* Each packaged reel shall be lagged with a layer of lumber in such manner that the reel shall be protected against damage.

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NOTE: FIT PULLEYS WITH SUITABLE BALL OR ROLLER BEARINGS.
"B" SHOULD BE BORED AND GROUND TO LIGHT PRESS FIT FOR BEARINGS.
"A" BORE OF BEARING AS RECEIVED.
MATERIAL: TOOL STEEL.
HEAT TREAT: HARDEN TO ROCKWELL C60 MINIMUM.

DIMENSIONS OF STEEL PULLEYS USED IN ENDURANCE TESTS

Cable Diameter	Pulley Ratio 1/	Pulley Dimensions				
		C ±1/64	D $^{+.005}_{-.000}$	E_R $^{+.002}_{-.000}$	F_R	G ±1/64
Inch		Inches	Inches	Inch	Inch	Inch
1/8	9.5	1-9/16	1.187	0.070	1/16	3/8
5/32	9.5	2	1.484	.086	1/16	3/8
3/16	9.5	2-3/8	1.781	.103	1/16	1/2
1/4	9.5	3-1/8	2.375	.133	1/16	1/2

1/ RATIO OF PULLEY DIAMETER "D" TO NOMINAL CABLE DIAMETER.

FIGURE 2.—*Endurance Test Pulley.*

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TABLE III.—*Reels for Cable*

Diameter of cable (inch)	Diameter of head	Traverse or distance between heads	Diameter of drum	Diameter of arbor hole	Diameter of head	Traverse or distance between heads	Diameter of drum	Diameter of arbor hole
	1,000 feet				3,000 feet			
	Inches	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*
1/32	12	4	8	1 1/8	12	4	8	1 1/8
1/16	12	4	8	1 1/8	12	4	8	1 1/8
5/64	12	4	8	1 1/8	16	4	10	1 1/8
3/32	12	4	8	1 1/8	16	4	10	1 1/8
7/64	16	4	10	1 1/8	16	7	12	1 1/8
1/8	16	4	10	1 1/8	16	7	12	1 1/8
9/64	16	7	12	1 1/8	16	10	8	1 1/8
5/32	16	7	12	1 1/8	16	10	8	1 1/8
3/16	18	7	12	2 1/8	18	10	8	2 1/8
7/32	18	7	12	2 1/8	18	10	8	2 1/8
1/4	18	10	10	2 1/8	24	10	10	2 1/8
5/16	18	10	10	2 1/8	24	10	10	2 1/8
11/32	18	10	8	2 1/8	32	16	16	2 1/8
3/8	18	10	8	2 1/8	32	16	16	2 1/8
7/16	24	12	10	2 1/2	50	12	26	5
1/2	24	14	10	2 1/2	50	12	26	5
9/16	26	14	11	2 1/2	50	14	30	5
5/8	32	18	18	3	50	16	30	5
3/4	32	22	18	3	50	22	30	5
7/8	50	12	30	5	50	30	30	5
1	50	14	30	5	60	24	30	5
	5,000 feet				10,000 feet			
	Inches	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*
1/32	12	4	8	1 1/8	16	4	10	1 1/8
1/16	16	4	10	1 1/8	16	7	12	1 1/8
5/64	16	7	12	1 1/8	16	10	8	1 1/8
3/32	16	7	12	1 1/8	16	10	8	1 1/8
7/64	16	10	8	1 1/8	18	10	8	1 1/8
1/8	16	10	8	1 1/8	24	10	10	1 1/8
9/64	24	10	10	1 1/8	24	16	10	2 1/8
5/32	24	10	10	1 1/8	24	16	10	2 1/8
3/16	24	10	10	2 1/8	24	16	10	2 1/8
7/32	24	10	10	2 1/8	32	20	16	3 1/8
1/4	32	18	16	2 1/8	36	22	18	3 1/8
5/16	32	18	16	2 1/8	36	22	18	3 1/8
11/32	32	20	16	3 1/8	50	16	26	3 1/8
3/8	32	20	16	3 1/8	50	16	26	3 1/8
7/16	50	14	26	5	50	26	26	5
1/2	50	16	26	5	50	28	26	5
9/16	50	20	30	5	60	22	30	5
5/8	50	26	30	5	60	30	30	5
3/4	60	24	30	5	66	34	30	5
7/8	60	28	30	5	68	38	30	5
1	60	36	30	5	76	40	30	5

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The lumber shall conform to Specification MM–L–736 or MM–L–751.

5.2.2 *Level C.* For domestic shipment, no additional packing is necessary.

5.3 **Marking of shipments.** The reels shall be marked in accordance with Standard MIL–STD–129.

6. NOTES

6.1 **Intended use.** The cable covered by this specification is intended for general aircraft use.

6.2 **Ordering data.** Procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Size and quantity of cable to be furnished.

(*c*) Where minimum lengths greater than 1,000 feet are required, or where lengths between 500 and 1,000 feet are not desired, the lengths to be furnished shall be specified.

(*d*) Selection of applicable levels of preservation, packaging, and packing.

6.3 **Definitions.**

6.3.1 *Wire.* Each individual cylindrical steel rod or thread is designated as a wire.

6.3.2 *Strand.* Each group of wires helically twisted or laid together is designated as a strand.

6.3.3 *Cable or wire rope.* A group of strands helically twisted or laid about a central core is designated as a cable. The strands and the core act as a unit.

6.3.4 *Preformed type.* Cable consisting of wires and strands shaped, prior to fabrication of the cable to conform to the form or curvature which they take in the finished cable is designated as preformed type.

6.3.5 *Diameter.* The diameter of cable is the diameter of the circumscribed circle.

6.3.6. *Lay or twist.* The helical form taken by the wires in the strand and by the strands in the cable is characterized as the lay or twist of the strand or cable, respectively. In a right-hand lay the wires or strands are in the same direction as the thread on a right-hand screw, and for a left-hand lay they are in the opposite direction.

6.3.7 *Pitch (or length of lay).* The distance, parallel to the axis of the strand or cable, in which a wire or strand makes one complete turn about that axis is designated as the pitch (or length of lay) of the strand or cable, respectively.

6.3.8 *Wire center.* The center of all strands is an individual wire and is designated as a wire center.

6.3.9 *Strand center.* A strand center consists of a single straight strand made of preformed wires, similar to the other strands comprising the cable in arrangement and number of wires.

6.4 **Cable lubricant.** A satisfactory aircraft control cable lubricant will reduce the internal friction and prevent corrosion. It will remain plastic at −67° F. and will not flow out of the cable at temperatures below 160° F. To insure a satisfactory service life, the lubricant should be resistant to oxidation. A satisfactory lubricant is described in NDRC Report entitled "The Corrosion Fatigue Failure of Aircraft Control Cables (N–101) OSRD No. 4819, Serial No. M–467," dated 26 February 1945. Copies of this document may be obtained from the Technical Data Division, Attention: TD–41, Bureau of Aeronautics, Navy Department, Washington 25, D.C.

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